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Software Requirements Specification (SRS) per IEEE 830 Standard

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for a web-based Learning Management System (LMS). The LMS aims to facilitate student learning by providing an online platform where students can enroll in courses, access learning materials, submit assignments, receive grades, and track their progress.

1.2 Scope

This document outlines the functional requirements for the LMS core features, user interface, performance, security, and data protection. External systems and tools are not within the scope of this document.

1.3 Definitions, Acronyms, and Abbreviations

- ANSI: American National Standards Institute
- API: Application Programming Interface
- LMS: Learning Management System
- SRS: Software Requirements Specification

1.4 References

IEEE STD 830-1984, "IEEE Recommended Practice for Software Requirements Specifications"

1.5 Overview

This document describes the product, its functions, user characteristics, constraints, assumptions and dependencies, external interfaces, system constraints, and assumptions and dependencies of the LMS.

2. Overall Description

2.1 Product Perspective

The LMS is a web-based application designed to support educational institutions in managing and delivering course content online.

2.2 Product Functions

- User Authentication (student, instructor, admin)
- Course Management (enrollment, creation, updating, deletion)
- Content Delivery (videos, documents, assignments, quizzes)
- Grading and Feedback Mechanism
- Email Notifications

- Progress Tracking and Reporting

2.3 User Characteristics

- Students: Individuals who enroll in courses and access learning materials.
- Instructors: Educators responsible for creating, updating, and deleting courses and providing assignments and quizzes.
- Admin Users: Personnel managing user accounts and courses.

2.4 Constraints

- The system shall support at least 500 concurrent users.
- All user data must be securely stored and protected.
- **The system shall respond to search queries within 2 seconds under normal load. [AI-SUGGESTED]**

2.5 Assumptions and Dependencies

- Stable internet connection is required for optimal usage of the LMS.
- Users have access to a web browser that supports HTML, CSS, and JavaScript.
- The system shall securely communicate using standard web protocols (HTTPS, SMTP).
- The LMS assumes that users have access to a stable internet connection and an appropriate web browser for optimal usage.
- The LMS relies on third-party tools for email communication, ensuring compatibility with SMTP protocols.

3. Functional Requirements

- REQ-001: The system shall build a web-based Learning Management System.
- REQ-002: Students should be able to register using their email addresses and create passwords (clarified for clarity).
- REQ-003: Students should be able to enroll in courses available for registration (consistency improved with other requirements).
- REQ-004: Students should access learning materials (videos, documents) within their enrolled courses (clarified for clarity).
- REQ-005: Instructors should be able to create, update, and delete courses as needed (consistency improved with other requirements).
- REQ-006: Instructors should upload assignments and quizzes for students to complete (clarified for clarity).
- REQ-007: Students should submit assignments and receive grades from instructors (consistency improved with other requirements).
- REQ-008: The system shall send email notifications for course deadlines and announcements (consistency improved with other requirements).
- REQ-009: Admin users should manage user accounts and courses as required (clarified for clarity).
- REQ-010: The system shall track student progress and generate performance reports (clarified for clarity).

4. Non-Functional Requirements

4.1 Performance Requirements

- The system shall respond to search queries within 2 seconds under normal load. [AI-SUGGESTED]

4.2 Security Requirements

- The system shall securely store and protect all user data, complying with relevant privacy laws and regulations.

4.3 Usability Requirements

- The system shall provide an intuitive and user-friendly interface for students, instructors, and admin users (consistency improved with other requirements).

4.4 Reliability Requirements

- The system shall maintain an uptime of 99.9%.

4.5 Maintainability Requirements

- The system shall be designed to facilitate maintenance and updates with minimal downtime.

4.6 Portability Requirements

- The system shall be compatible with various web browsers and devices (consistency improved with other requirements).

5. External Interface Requirements

5.1 User Interfaces

- The LMS shall have a web-based user interface accessible via standard web browsers.

5.2 Hardware Interfaces

- No specific hardware interfaces are required for the LMS.

5.3 Software Interfaces

- The LMS shall support API integrations with other educational tools and systems as necessary (clarified for clarity).

5.4 Communication Interfaces

- The LMS shall communicate securely using standard web protocols (HTTPS, SMTP) (consistency improved with other requirements).

6. System Constraints

- The system must operate within the constraints defined by the hardware and network infrastructure provided.

7. Assumptions and Dependencies

- The LMS assumes that users have access to a stable internet connection and an appropriate web browser for optimal usage.

- The LMS relies on third-party tools for email communication, ensuring compatibility with SMTP protocols.

8. Verification and Validation (V&V) Plan Development: (not included as part of the current SRS but to be developed separately based on the improved requirements)
9. Review and Approval: (to be conducted after the V&V plan is developed and approved)